Moises Carrera

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PROFESSIONAL SUMMARY

Mechatronics Engineer and ESTEEM Graduate Student (Notre Dame '26) with experience in hardware-software integration, multi-sensor systems, and PCB design, combined with customer-facing experience. Proven ability to translate complex engineering findings into actionable decisions for cross-functional teams. Entrepreneurial mindset developed through strategy, product commercialization, and a licensed consumer startup, bridging engineering expertise with customer development. Bilingual in Spanish and English.

EDUCATION

University of Notre Dame — South Bend, IN
MS in Engineering, Science, Technology and Entrepreneurship (ESTEEM) — May 2026

Anahuac Queretaro University — Queretaro, Mexico
BEng in Mechatronics Engineering | GPA 3.72/ 4.0 | Minor: Strategic Management of the Company — May 2025

Università Degli Studi di Firenze — Florence, Italy
Academic Exchange "Ingegneria Gestionale" (Management Engineering) — January 2024

- Completed coursework in LEAN, TRIZ innovation, problem-solving methodology, and analytic tools for business

EMPLOYMENT

Irish Nostalgia — South Bend, IN
Product Growth & E-Commerce Lead — October 2025 - Present

- Managed end-to-end product development and quality assessment for a 3D-printed and laser-cut licensed memorabilia line, overseeing materials selection, Bambu Lab equipment operation, and production output at scale.
- Managed 7+ institutional B2B contracts establishing an outreach-to-close motion for high-value accounts.
- Rebuild the Shopify storefront end-to-end, redesigning the customer funnel grounded in VOC data points and mentors.

Tycho.AI — South Bend, IN
Product Management Advisor - Graduate Student — July 2025 - Present

- Conducted 50 primary technical and customer discovery interviews evaluating performance requirements, failure modes, and adoption barriers for autonomous drone navigation systems in commercial environments.
- Developed a multi-year Go to Market strategy and a financial model to guide commercial launch sequencing for a venture-backed ag-tech startup.

Mabe TyP — Queretaro, Mexico
NTI Intern - Artificial Vision & UX Research — March 2024 - March 2025

- Led research and usability testing for a vision-based refrigerator system, identifying friction points and implementing data-driven design changes that improved system usability by 50%.
- Designed two 2-layer production PCBs in Altium Designer for multi-sensor IoT refrigerator units; selected component footprints, applied design rules, and adapted hardware reference designs from senior engineers into fabrication-ready layouts.
- Prepared and presented technical reports to cross-functional teams, translating complex engineering findings into actionable design improvements.

LEADERSHIP & PROJECTS

Innova - Leadership in Technological Innovation — Queretaro, Mexico
Resource Planning President — August 2022 – December 2024

- Completed a competitive leadership diploma focused on innovation, technology strategy, and organizational decision-making.
- Developed management and cross-functional team collaboration skills through applied leadership activities and real-world case challenges.

Anahuac Racing Team — Queretaro, Mexico
Suspension Engineer — June 2021 – May 2023

- Designed and manufactured the full suspension system for a nationally competing single-seat off-road vehicle (BAJA SAE 2022), applying SolidWorks for modeling and simulation.
- Delivered on-site mechanical support and rapid troubleshooting during competition events under high-pressure conditions.

SKILLS & CERTIFICATIONS

Languages: Spanish (Native), English (Advanced), Italian (Basic conversational).
Engineering tools: SolidWorks (CSWA Certified), Altium Designer, Python, Rapid Prototyping, Bambu Lab 3D Printing.
Technical Skills: Hardware-Software Integration, PLC Keyence (Ladder Logic, basic), Agile Methodologies, CNC (G-code, basic)
Certifications: CSWA SolidWorks Associate (2025) | Data Science & AI Basic Level (2025) | Leadership in Technological Innovation GE Vernova (2024).